From Observation to Action: Evaluating Hypothesis-Driven Rule Discovery in Unseen Interactive Environments

Short title: Interactive Rule Discovery under Distribution Shift

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Abstract

Generalising to environments whose rules were never observed is a central open problem for learning agents. We ask whether an agent that explicitly forms, tests and revises hypotheses about hidden environmental rules generalises better than reactive exploration. We introduce a CPU-only, procedurally generated grid-world benchmark with compositional hidden rules (key–door, switch–bridge, push-block) and five held-out generalisation regimes, and compare a random agent, a reactive heuristic agent, and a hypothesis-driven agent that selects experiments by a utility combining expected progress, information gain and cost. Across 30 independent seeds × 30 environments per condition, at difficulty 3 the hypothesis-driven agent succeeds in 97.3% of episodes versus 38.1% for the reactive agent and 0.0% for random; it also generalises better to unseen rule compositions (held-out success 0.97–1.00 across five regimes vs 0.40–0.83). Ablations show that evidence persistence, hypothesis revision and choosing experiments by expected value each matter, whereas the explicit information-gain weight λ had no significant effect. Two findings temper the headline: cross-episode priors gave no measurable benefit and produced negative transfer when the learned relation was reversed, and the agent's advantage is partly attributable to inductive biases that match the generator. The study is synthetic, small-scale and deterministic, and makes no claim about ARC-AGI-3 or general intelligence.

Keywords: interactive rule discovery, hypothesis testing, active experimentation, compositional generalisation, procedural benchmarks.

1. Introduction

Learning systems tend to perform well on distributions they have seen and poorly on structures they have not. Interactive settings sharpen this problem: an agent receives observations and the consequences of its own actions, but never an explicit description of the rules that govern the world. To succeed it must *infer* those rules from interaction — which key opens which door, which switch changes which object, which obstacles must be pushed — and reuse the inference when the rules recombine in new ways. This mirrors the broader aim of measuring skill-acquisition efficiency rather than skill on a fixed task distribution [Chollet 2019].

A common route to such behaviour is end-to-end reinforcement learning, which can need many interactions and often generalises poorly across held-out level structures [Cobbe et al. 2020; Kirk et al. 2023]. An alternative, rooted in work on scientific reasoning [Klahr & Dunbar 1988] and "theory-like" cognition [Gopnik & Meltzoff 1997; Tenenbaum et al. 2011], is to maintain *explicit hypotheses*, choose experiments that discriminate between them, and revise them on contradiction. Whether this explicit mechanism, *by itself*, buys sample efficiency and generalisation is an empirical question we study in a controlled, fully inspectable setting.

Research question. Can explicit hypothesis formation, testing and revision improve an agent's ability to discover hidden rules and generalise to previously unseen interactive environments compared with random and purely reactive exploration?

We pre-specify four hypotheses and design experiments able to falsify them: **H1** the hypothesis-driven agent has higher success and/or fewer interactions on unseen environments than a reactive agent; **H2** it generalises better to unseen *combinations* of known rule types; **H3** it recovers from incorrect hypotheses more efficiently than an agent without explicit revision; **H4** its advantage grows with rule complexity.

Scope. This work is an independent, controlled benchmark *inspired by* the problem of interactive rule discovery; it does **not** reproduce ARC-AGI-3 or any other external benchmark, uses no language models, and makes no claim of general or human-level reasoning.

Contributions.

1. A CPU-efficient procedural benchmark for hidden-rule discovery with five train/test generalisation regimes (interpolation, compositional, structural, rule-permutation, novel combination).
2. A hypothesis-driven interactive agent with explicit confidence updating, contradiction-based rejection, episodic memory and an information-gain-weighted experiment selector.
3. A controlled comparison against random and reactive baselines with 30 seeds, bootstrap confidence intervals, Welch/Mann–Whitney tests with Holm correction, and effect sizes.
4. An empirical study of compositional generalisation, active experimentation, recovery from incorrect hypotheses and failure modes — including two degenerate behaviours of our own agent found and fixed during development.

2. Related Work

Generalisation in RL. Procedurally generated benchmarks such as Procgen [Cobbe et al. 2020] and grid-world suites like BabyAI and Minigrid [Chevalier-Boisvert et al. 2019; 2023] expose the gap between training-level and held-out-level performance; Kirk et al. [2023] survey zero-shot generalisation in deep RL. Our environments are in the same spirit but are designed so that *rule*

bindings, rather than layouts, are what must be discovered. **Compositional generalisation** has been studied mainly in language [Lake & Baroni 2018; Keysers et al. 2020]; we adapt the idea to rule compositions (train on pairs, test on triples).

Exploration and active learning. Count-based and prediction-error intrinsic motivation [Bellemare et al. 2016; Pathak et al. 2017], information-gain exploration [Houthooft et al. 2016] and active learning [Settles 2009] motivate our utility $U(a)=R+\lambda\ IG-\mu\ c$; unlike these, our information gain is computed over a small *symbolic* hypothesis space. **Model-based and Bayesian RL** [Sutton & Barto 2018; Ghavamzadeh et al. 2015; Moerland et al. 2023] maintain beliefs about dynamics; our manager is a deliberately simple, interpretable special case. **Meta-RL** approaches learn to adapt from experience [Duan et al. 2016; Wang et al. 2016; Finn et al. 2017; Zintgraf et al. 2020] and are a natural, un-tested comparison class here.

Reasoning benchmarks and cognition. ARC-style benchmarks stress inference of unseen rules from few examples [Chollet 2019]; interactive variants are the motivation for this study, not its claim. Human-inspired accounts of scientific discovery as dual-space search over hypotheses and experiments [Klahr & Dunbar 1988] and of learning as theory revision [Gopnik & Meltzoff 1997; Tenenbaum et al. 2011] inspired the agent's architecture. For statistical practice we follow recommendations on seeds and uncertainty in RL evaluation [Henderson et al. 2018; Colas et al. 2018; Agarwal et al. 2021].

3. Problem Formulation

An environment is $E=(S,A,T,R,O,G)$: states S, actions A, transition function T, reward/feedback R, observation function O, and a hidden rule structure G. The agent observes $o_t=O(s_t)$ and feedback but never G. Let $\mathcal{H}=\{h_1,\ldots,h_n\}$ be a hypothesis space of candidate rule bindings with confidences $c_t(h)$. At time t the agent holds beliefs $\mathcal{H}_t$, selects a_t, receives o_{t+1} and feedback, and updates $c_{t+1}(h)$ for each hypothesis whose prediction the outcome tests. An episode is successful if the target is reached within $T_{max}=200$ steps.

4. Methodology

4.1 Environment and hidden rules

Grids are $n\times n$ ($n\in\{8,10,12,14\}$) with walls, an agent, a target, keys, doors, switches, chasms (impassable until bridged), movable blocks and traps. Actions: UP, DOWN, LEFT, RIGHT, INTERACT, PUSH, WAIT. Observations are fully observable structured tuples (grid, agent position, facing, held key, visible objects with colour/state, feedback string, reward, done); the hidden rule object is never exposed (verified by tests). Three **gate rules** can be composed: *KEY_DOOR* (a key of colour k opens the door of colour $\phi(k)$; the agent carries **one** key at a time, and picking up another swaps), *SWITCH_BRIDGE* (a one-shot switch of colour s activates the chasm of colour $\psi(s)$), and *PUSH_BLOCK* (a block in a gap must be pushed twice). Distractor keys/switches (no matching gate) and lethal traps are added with difficulty. The bindings ϕ,ψ are hidden and differ per environment, so they cannot be memorised and must be discovered.

4.2 Procedural generation and solvability

`EnvironmentGenerator` places k vertical walls with one gate each; the key/switch for gate i is placed in zones $\le i$, guaranteeing a sequential dependency chain switch → key → door → … → target. Every instance is verified solvable by a closure-based solver (including block final positions); unsolvable draws are resampled deterministically from the seed. Colour mappings can be `random`, `identity` (key colour = door colour) or `anti` (never equal), which Experiments D and the recovery study use. Difficulty 1–5 increases grid size (8–14), number of gates (1–5), distractors and traps (Table 1).

Table 1. Environment configurations per difficulty level (KE=key-door, SW=switch-bridge, PU=push-block).

Difficulty	Grid	Gates	Distractor_keys_switches	Traps
1	8	KE	0	0
2	10	KE+SW	0	1
3	10	KE+SW+PU	1	2
4	12	KE+SW+PU+KE	2	3
5	14	KE+SW+PU+KE+SW	3	4

4.3 Agents

Random samples uniformly from the seven actions. **Reactive** uses the current observation and BFS heuristics (go to target if a path exists; pick up keys when empty-handed; toggle switches; interact with a door in front when holding a key; push a block in front) plus one *lose-shift bit* that makes it swap keys after a failed door attempt. It keeps no record of which key/door pairs it tried. **Hypothesis-driven** is described next. Both non-random agents avoid traps and know the action semantics (INTERACT picks up/toggles/opens, PUSH pushes); what they must discover are the *bindings*.

Table 2. Agent architectures.

Agent	Policy	Memory	Hypotheses	Transfer
Random	uniform over 7 actions	none	none	none
Reacti	BFS heuristics + lose-shift bit	none (1 bit)	none	none

Agent	Policy	Memory	Hypotheses	Transfer
ve				
Hypothesis	macro-actions chosen by U = R + lam*IG* - *mu*cost	episodic (unbounded)	KD/SB/PUSH binding hypotheses, alpha/beta updates, rejection	optional cross-episode priors

4.4 Hypothesis-driven agent

Representation. A hypothesis is `Hypothesis(rule_type, subject, condition, action, predicted_effect, confidence, evidence_count, contradiction_count)`, e.g. (*KD*, red key, blue door). The space contains one hypothesis per (key colour, door colour), (switch colour, chasm colour) and one for "blocks can be pushed". *Lifecycle.* Hypotheses are created from the observation, assigned an initial confidence (0.5 without priors), tested by an experiment, and updated by `update_hypothesis`: a correct prediction adds $\alpha=0.6$ to confidence (clipped to 1) and marks the hypothesis confirmed; an incorrect one subtracts $\beta=0.6$, increments the contradiction count, and rejects the hypothesis after two contradictions. Decisions use probabilities $p(k \to d)$ obtained by normalising confidences over non-rejected candidates, with mutual exclusion (a confirmed pair eliminates alternatives for that key and door). *Memory.* Each tested hypothesis is stored as `MemoryEntry(observation, action, outcome, hypothesis, prediction, error)`. Hypothesis state is *recomputed* from priors plus the memory contents at each decision, so memory capacity (unbounded / last-3 / none) directly controls what the agent can remember. *Active experimentation.* Decisions are over macro-actions (go to target; pick/swap a key; test held key on a reachable door; toggle a switch; push a block). Each has expected task reward $R=V\cdot p$ (value $V=3$ per gate, 10 for the target), information gain *IG* (binary entropy of the tested hypothesis; for a switch, summed over unresolved chasms), and cost *c* (path length). The agent maximises

$U(a)=R(a)+\lambda\, IG(a)-\mu\, c(a)$, $\lambda=0.5$, $\mu=0.05$ (default).

Macros with negligible value ($R<0.1$ and $IG<0.2$) are discarded, and key macros only count doors that are currently reachable. Exploration is a macro whose weighted information term exceeds its reward term. *Cross-episode priors* (optional, "transfer") count how often confirmed bindings were same-coloured and whether pushing worked, and use these as initial confidences in later episodes; they are frozen at test time.

Algorithm.

```
for each episode:  reset env; rebuild H from priors + memory M
  while not done:
    o <- observe;  H <- rebuild(M)
    candidates <- {macro-actions available in o}
    for a in candidates: R(a)=V*p(success), IG(a)=entropy, c(a)=path length
    drop negligible-value candidates;  a* <- argmax R + lam*IG - mu*c
    execute a*; on its terminal action observe outcome
    for each tested h: error <- (predicted != observed); store MemoryEntry
in M
       update_hypothesis(h, not error)   # +alpha / -beta, reject after 2
contradictions
  record metrics;  (if transfer) update priors from confirmed bindings
```

4.5 Train/test regimes

Test environments are held out by construction (disjoint seed ranges and, for A–E, disjoint compositions, sizes or colour sets; verified by `test_generalization.py`). **A (interpolation):** train on single rules {KD},{SB},{PB}; test on pairs. **B (compositional):** train on the three pairs; test on the triple KD+SB+PB. **C (structural):** train 10×10, test 14×14, same rules. **D (rule permutation):** train on identity-mapped rules with colours {red, blue, green}; test on identity-mapped rules with disjoint colours {yellow, purple, orange}. **E (novel combination):** train on {KD} and {SB+PB}; test KD+SB+PB. Learning agents (the transfer variant) train for 40 episodes per seed and are then frozen; non-learning agents are simply evaluated on both splits. The *generalisation gap* is held-out-train success minus test success.

4.6 Evaluation protocol

All experiments use 30 independent seeds, each evaluating 30 fresh environments per condition (900 episodes per cell). The seed is the unit of analysis. We report mean, standard deviation, 95% percentile-bootstrap CIs over seeds [Efron & Tibshirani 1993], Welch's *t*-test [Welch 1947], the Mann–Whitney *U* test, Cohen's *d* [Cohen 1988], and Holm correction [Holm 1979] across all 42 reported tests. Metrics: success rate, interactions to success, sample efficiency (successes per interaction), hypothesis accuracy (fraction of tested hypotheses that are true under the hidden rules), recovery rate (fraction of gates with a failed attempt that were later opened), generated/rejected/retained hypotheses, contradictions, redundant experiments, exploration share, CPU time, actions/s and peak memory. Because steps-to-success is computed only over successes, it is subject to survivor bias and is interpreted alongside success rate.

5. Results

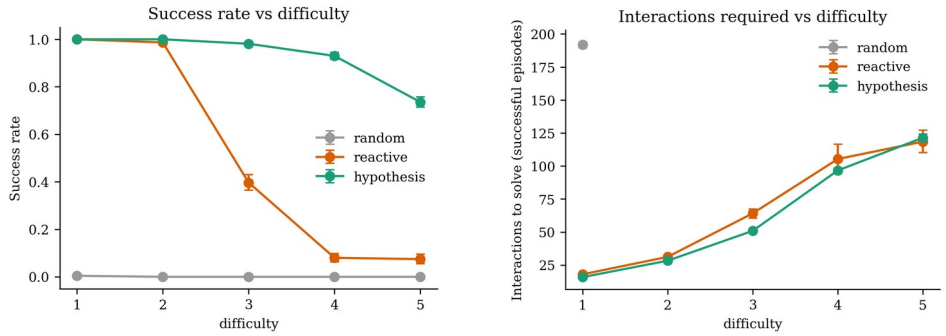
All numbers below are produced by `scripts/run_all.py` and traced to `data/results/*.csv`; full tables are in the supplement.

5.1 Main comparison and difficulty scaling (H1, H4)

Table 3. Main results, difficulty 3 (mean [95% bootstrap CI over seeds]).

Agent	Success rate	Steps to success	Success/interaction x1e3	Actions/s	Hyp. accuracy	Recovery rate
random	0.000 [0.000, 0.000]	n/a	0.00 [0.00, 0.00]	94575	-	0.083 [0.046, 0.123]
reactive	0.381 [0.347, 0.419]	59.8 [56.3, 63.3]	2.87 [2.51, 3.25]	12361	-	0.277 [0.258, 0.295]
hypothesis	0.973 [0.961, 0.983]	50.5 [49.6, 51.4]	18.63 [17.93, 19.26]	26282	0.887 [0.848, 0.921]	0.992 [0.985, 0.998]

At difficulty 3 the hypothesis-driven agent reaches the target in 97.3% of episodes versus 38.1% (reactive) and 0.0% (random); the difference to the reactive agent is +0.584 (Welch $p < 0.001$ after Holm, Cohen's $d = 8.7$). Successful hypothesis-agent episodes also use fewer interactions (50.5 vs 59.8). Tested hypotheses were true 88.7% of the time; the agent generated 5.0 hypotheses and recorded 1.15 contradictions per episode, and spent 0.0% of its decisions on exploration.



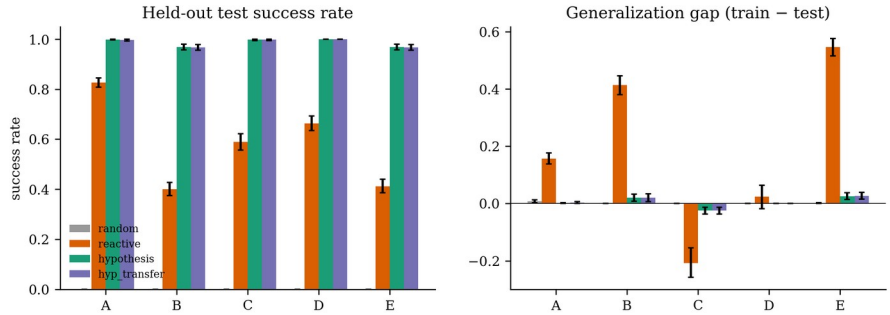
Across difficulties (Fig. 4) the hypothesis agent's success is 1.00, 1.00, 0.98, 0.93, 0.74 for $d=1..5$, versus 1.00, 0.99, 0.40, 0.08, 0.07 for the reactive agent. The *advantage* (+0.00, +0.01, +0.58, +0.85, +0.66) is zero at $d=1$ (both ceilinged) and grows through $d=4$, **but it is not monotonic**: at the highest difficulty the hypothesis agent itself degrades and the gap narrows. **H4 is therefore only partly supported:** the advantage grows with complexity up to a point, then shrinks. Steps-to-success differences are significant at $d \leq 3$ (Holm $p < 0.001$ at $d=3$) but not at $d=4$ ($p = 1.000$) or $d=5$ ($p = 1.000$), where few reactive episodes succeed and those are easy ones (survivor bias). Random exploration is essentially never successful beyond $d=1$.

H1 is supported for success rate at $d \geq 2$ in this benchmark, and for interactions at $d \leq 3$.

5.2 Generalisation to unseen environments (H2)

Table 4. Held-out test success (generalisation gap = train − test), mean over seeds. Full CIs: Table S2b.

Regime	random	reactive	hypothesis	hyp_transfer
A	0.00 (gap +0.01)	0.83 (gap +0.16)	1.00 (gap +0.00)	1.00 (gap +0.00)
B	0.00 (gap +0.00)	0.40 (gap +0.41)	0.97 (gap +0.02)	0.97 (gap +0.02)
C	0.00 (gap +0.00)	0.59 (gap -0.21)	1.00 (gap -0.02)	1.00 (gap -0.02)
D	0.00 (gap +0.00)	0.66 (gap +0.02)	1.00 (gap +0.00)	1.00 (gap +0.00)
E	0.00 (gap +0.00)	0.41 (gap +0.55)	0.97 (gap +0.03)	0.97 (gap +0.03)



Held-out success and generalisation gap by regime.

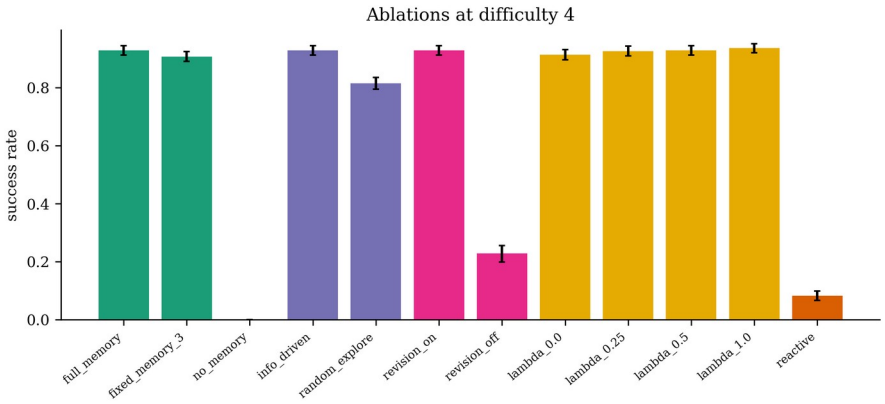
The hypothesis agent's held-out success ranges 0.969–1.000, and exceeds the reactive agent's in every regime. In the *compositional* regimes (A, B, E) the reactive agent loses substantially when rules are recombined (gaps 0.16, 0.41, 0.55) while the hypothesis agent's gap is 0.001, 0.020, 0.026. Gaps conflate the intrinsic difficulty of the test distribution with failure to transfer, so they are most informative when compared *between* agents on the same split. **H2 is supported for compositional recombination.** Two

regimes need caution. In **C (structural)** both agents score *higher* on the larger grid (negative gaps) because larger grids offer more space around gates, so size is confounded with difficulty and C does not cleanly test structural generalisation. In **D (rule permutation)** the hypothesis agent is at ceiling (1.000) with or without cross-episode learning, so D cannot distinguish "learned the relation" from "tests bindings from scratch". Importantly, **the transfer variant never outperformed the plain hypothesis agent** (all Holm-corrected differences non-significant, Table S3): within-episode hypothesis testing already generalises here, and learned priors added nothing measurable.

5.3 Ablations

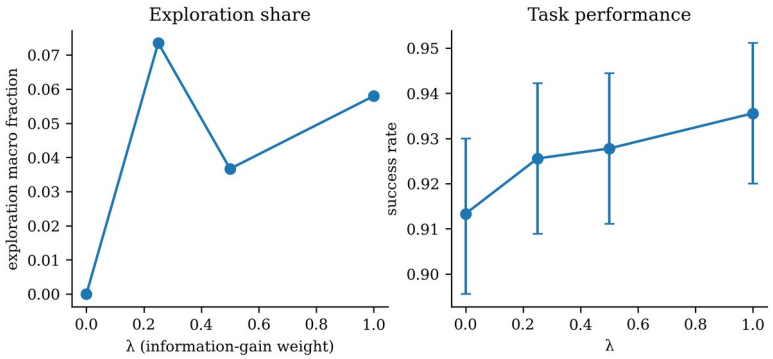
Table 5. Ablations at difficulty 4 (mean ± SD over seeds). Full CIs: Table S2.

Ablation	Variant	Success	Redundant tests
1_memory	full_memory	0.928 ± 0.047	0.8
1_memory	fixed_memory_3	0.907 ± 0.047	0.4
1_memory	no_memory	0.000 ± 0.000	154.9
2_active_exploration	info_driven	0.928 ± 0.047	0.8
2_active_exploration	random_explore	0.814 ± 0.058	0.0
3_revision	revision_on	0.928 ± 0.047	0.8
3_revision	revision_off	0.228 ± 0.081	116.5
4_info_gain_weight	lambda_0.0	0.913 ± 0.049	0.8
4_info_gain_weight	lambda_0.25	0.926 ± 0.048	0.8
4_info_gain_weight	lambda_0.5	0.928 ± 0.047	0.8
4_info_gain_weight	lambda_1.0	0.936 ± 0.044	0.8
reference	reactive	0.082 ± 0.046	-



Ablation results at difficulty 4.

Memory. Full memory (0.928) and a 3-entry fixed memory (0.907) are statistically indistinguishable (p = 1.000); with no memory the agent cannot retain contradictions, repeatedly re-tests the same pair, and almost never succeeds (0.000). This extreme result reflects our design (hypothesis state is derived from memory) and shows that persistent evidence is *necessary*, not that memory size is finely tuned. *Active experimentation.* Information-driven selection (0.928) far exceeds random exploration over feasible macros (0.814, d = 2.1). *Revision.* Disabling revision (failures are ignored) drops success from 0.928 to 0.228. *Information-gain weight* (Fig. 10): λ=0.0: 0.913, λ=0.25: 0.926, λ=0.5: 0.928, λ=1.0: 0.936. Success rises monotonically with λ, but the differences are small and **not significant** after Holm correction (λ=1 vs λ=0: Δ = +0.022, p = 1.000, d = 0.48); redundant tests were identical across λ (0.84) and the share of exploration macros was non-monotonic in λ (Fig. 10). The data therefore do **not** show that an explicit information bonus helps beyond expected-value selection: when a binding is uncertain, acting on it *is* the experiment, so λ=0 already tests hypotheses. The large effect of 'info_driven' over 'random_explore' reflects choosing among macros by *expected value*, not the λ term.



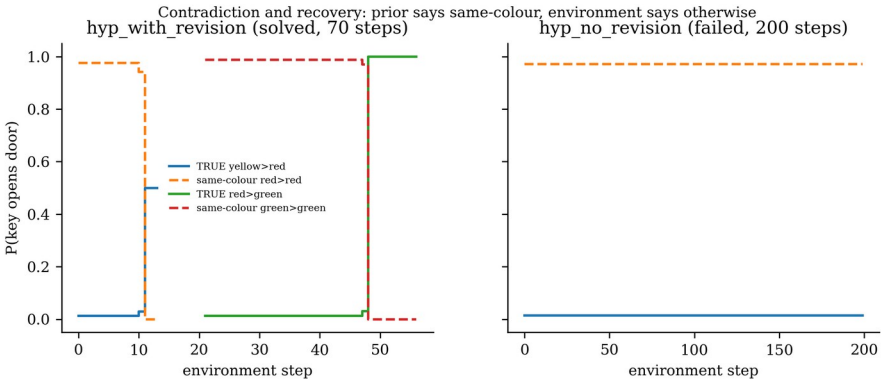
Exploration share and success vs information-gain weight.

5.4 Contradiction recovery (H3)

Agents pre-trained on identity-mapped rules (key colour = door colour) are tested on `anti` mappings in which that relation is always wrong, so the first hypothesis the prior favours is incorrect.

Contradiction-recovery experiment (anti-mapped test environments).

Agent	Success rate	Recovery rate	Failed tests/episode	Steps to success
reactive	0.667 [0.637, 0.694]	0.611 [0.592, 0.629]	2.09	89.8 [86.7, 92.9]
hyp_no_revision	0.179 [0.154, 0.203]	0.219 [0.191, 0.248]	0.98	40.7 [39.2, 42.4]
hyp_with_revision	0.780 [0.749, 0.808]	0.918 [0.903, 0.934]	1.33	61.0 [59.7, 62.3]
hyp_no_prior	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.06	53.1 [52.1, 54.2]



Hypothesis probability over time with and without revision (same held-out environment).

With revision enabled the agent recovers 91.8% of failed gates and succeeds in 78.0% of episodes; with revision disabled it recovers 21.9% and succeeds in 17.9% ($p < 0.001$, $d = 7.8$). **H3 is supported** with respect to revision. However, a more sobering result stands beside it: the revising agent with a *wrong* prior succeeds no more often than the reactive agent (0.780 vs 0.667; Holm $p < 0.001$), and the same agent *without* any prior succeeds in 100.0%. Misleading priors are costly even when revision works, i.e. we observed **negative transfer** from the cross-episode relation prior.

5.5 Computational cost

On a single CPU core the agents run at roughly 94575 (random), 12361 (reactive) and 26282 (hypothesis) actions/s, with 2.1 ms of CPU time per hypothesis-agent episode and a peak process size of about 27 MB. The full pipeline runs on a consumer CPU without GPUs or external services.

6. Failure Analysis

We classified the 236 failed hypothesis-agent episodes (of 900) at difficulty 5 into mutually exclusive categories by fixed rules (priority order: repeated refuted hypothesis; distractor interference = ≥25% of steps on macros involving keys/switches that no gate needs; long-horizon dependency = ≥4 gates; otherwise insufficient exploration). Counts: distractor interference: 189, long-horizon dependency: 45, repeated refuted hypothesis: 2. Success at difficulty 5 rises from 0.23 (100 steps) to 0.74 (200), 0.86 (400) and 0.87 (800 steps): extra budget helps but plateaus, so a residual of failures is *not* a budget problem. Examples are listed in the supplement. We note the other categories requested by our protocol: *ambiguous observation* cannot occur (full observability and deterministic feedback); *contradiction not detected* is exactly the revision-off condition of §5.3–5.4; *computational limitation* was not observed.

Development honesty note. Our first full run exposed two degenerate behaviours of the hypothesis agent: (i) re-testing an already-refuted pair indefinitely because a useless but cheap experiment had higher (less negative) utility than a useful but distant one, and

(ii) swapping two keys back and forth to collect an information-gain bonus for doors it could not reach. Both were fixed (negligible-value filter; reachable-door restriction) *before* the final runs; no pre-fix results appear in any table or figure here, and the diagnosis is recorded in the repository's `DEVLOG.md`. Utility-design hygiene of this kind is a practical finding in itself.

7. Discussion

Was H1 supported? Yes in this benchmark: higher success from $d=2$ and fewer interactions at $d \leq 3$. *Where does the agent succeed?* Wherever bindings must be *identified among several candidates* (several keys, several switches), and when rules are recombined. *Where does it fail?* In the longest dependency chains with many distractors (§6) and when a learned prior is wrong (§5.4). *Does active experimentation help?* Selecting experiments by expected value helps a great deal versus random choice among feasible macros (success 0.928 vs 0.814). The *explicit information-gain term* did not help significantly (λ=0: 0.913; λ=1: 0.936; p = 1.000 after Holm) and we cannot claim it contributes in this benchmark. *Do explicit hypotheses matter?* The revision-off and no-memory ablations show that the *update-and-remember* loop is what the agent's performance depends on. *Is generalisation limited?* Within this rule vocabulary it is not, but that is largely by design.

Important qualification. The hypothesis agent's templates (key ↔ door, switch ↔ chasm, push) and its mutual-exclusion assumption are aligned with how the generator builds worlds. The reactive baseline lacks both. The results therefore show that *an agent with the right structural prior and an evidence-tracking loop outperforms a stateless heuristic*, not that explicit hypotheses are superior to every learned or implicit alternative (e.g. recurrent meta-learners), which we did not test.

8. Limitations

The environments are synthetic, deterministic, fully observable and have a small action space and a limited rule vocabulary; the hypothesis space is hand-specified and the agent knows the action semantics, so we test *binding discovery*, not discovery of rule *types* or of action semantics. Utility constants (V,γ,μ,α,β) were set by hand and not tuned, and λ's best value lay on the edge of the grid. The reactive baseline is a reasonable but hand-written heuristic and could be improved; stronger baselines (tabular or neural RL, meta-RL) are future work. Regime C confounds size with difficulty and regime D is at ceiling. Steps-to-success is subject to survivor bias. Experiments are CPU-scale; per-seed variance is small, which makes effect sizes very large and p-values extreme, so effect sizes and CIs are more informative than p-values. We do not model noise/stochastic transitions, natural-language reasoning or robotics, nor compare with frontier foundation models. Nothing here supports claims of general intelligence.

9. Conclusion

In a controlled, CPU-scale benchmark of hidden-rule discovery, explicit hypothesis-driven interaction improved success (97.3% vs 38.1% at difficulty 3), interaction counts at low-to-moderate difficulty, and held-out performance on unseen rule *compositions* (reactive held-out success 0.40–0.83 vs 0.97–1.00). The benefit depends on remembering evidence, revising on contradiction and choosing experiments by expected value (the explicit information-gain bonus was not significant); cross-episode priors gave no benefit and could hurt. Whether these conclusions transfer to richer, stochastic or learned-hypothesis settings remains open.

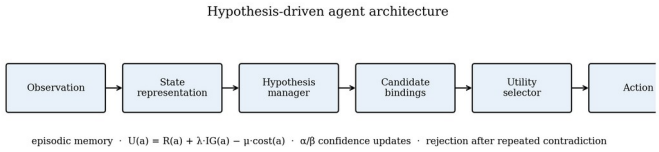
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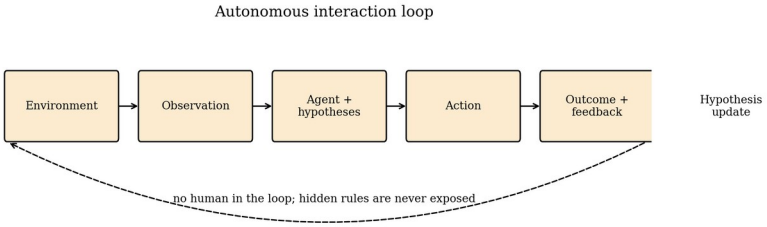
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Reference details should be verified against the original sources before submission.

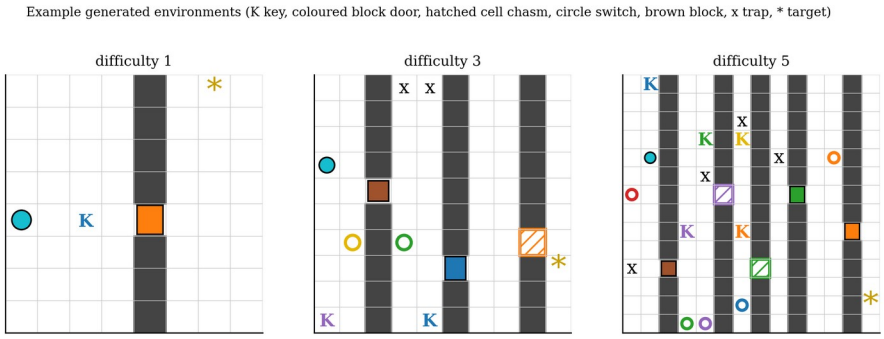
Figures of the system



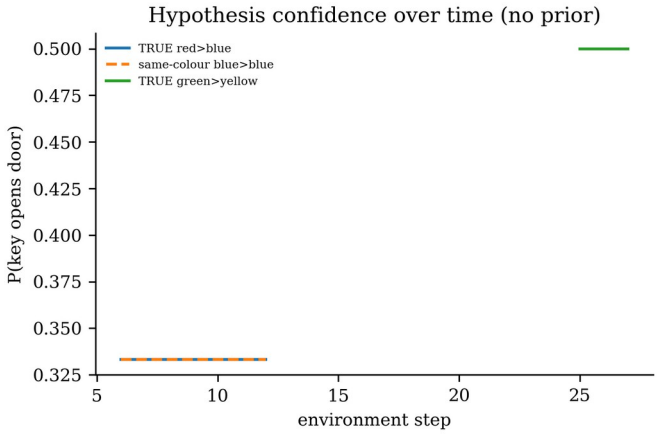
Agent architecture.



Interaction loop.



Example generated environments.



Hypothesis confidence over time (agent without prior; solid = true binding, dashed = same-colour hypothesis).